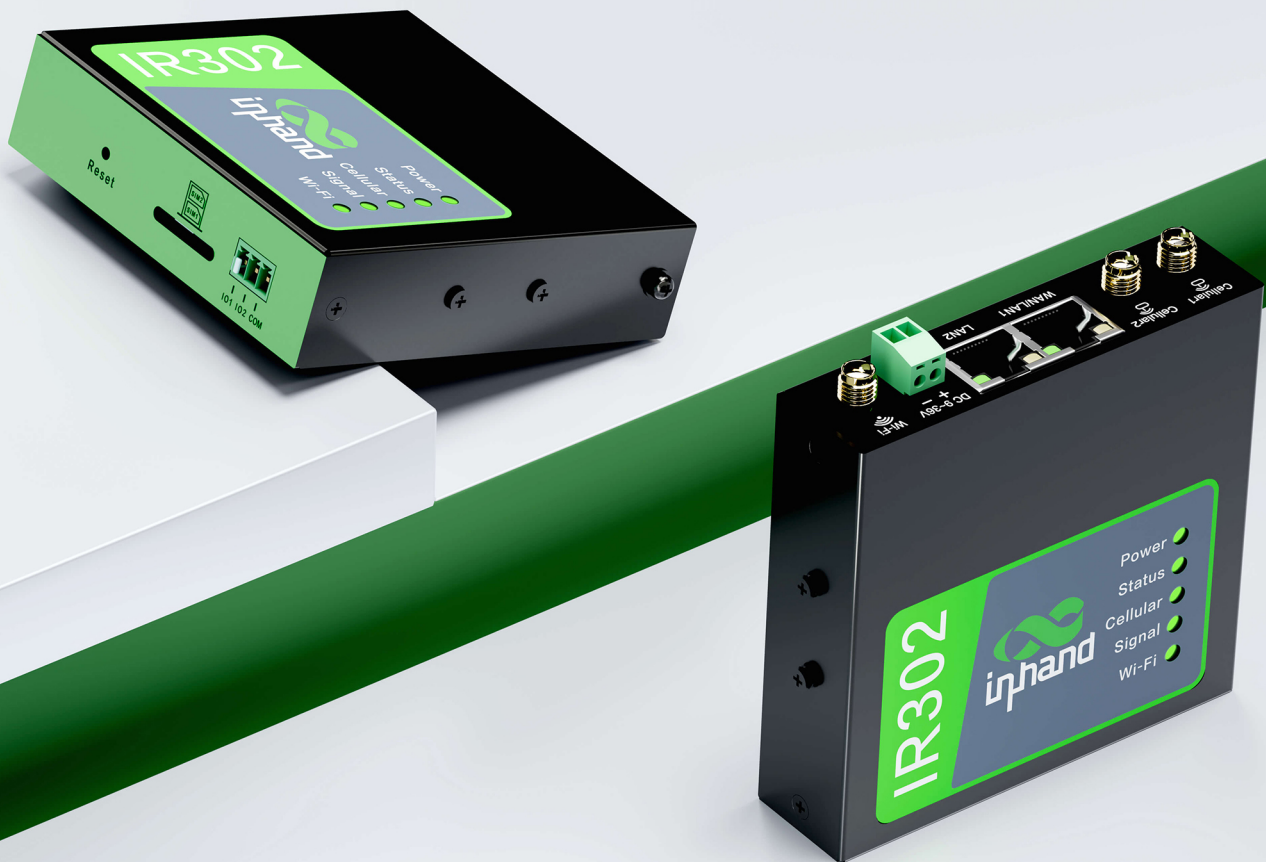


Compact Industrial LTE Router



InRouter302 Industrial Cellular Router

· 4G

· Wi-Fi

· Security

· Cloud-Managed

1. Product Overview

InRouter302 is a compact industrial cellular router that integrates 4G LTE, Wi-Fi, VPN, and cloud management for unattended remote sites.

Positioning: Cost-effective industrial router for secure connectivity, resilient networking, and remote O&M.

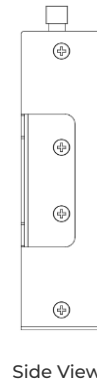
Key features:

- **Flexible access:** Cellular, wired, and Wi-Fi uplink options for diverse deployment scenarios
- **Reliable connectivity:** 4G/wired/Wi-Fi backup, dual SIM switching, VRRP, and multi-layer link detection
- **Security by design:** Firewall policies, ACL, VPN, encryption, and Wi-Fi security
- **Industrial-grade build:** Wide voltage input and -20~70°C operation for harsh environments
- **Cloud operations:** Device Manager platform for large-scale remote fleet management

Core Technical Specifications

Technical Item	Specification
Cellular Network	LTE Cat4/Cat1 (model-dependent), GSM/GPRS/EDGE, UMTS/HSPA+/EVDO/TD-SCDMA, TDD/FDD LTE
VPN & Data Security	PPTP/L2TP/GRE/IPSec/OpenVPN/WireGuard/ZeroTier, CA certificate support
Wi-Fi (Optional)	2.4GHz, IEEE 802.11 b/g/n, up to 150Mbps, AP/Client
Firewall & Access Control	SPI firewall, anti-DoS, ACL, URL filter, port mapping, virtual IP mapping, IP-MAC binding
Reliability	VRRP, heartbeat detection with auto-redial, dual SIM failover, embedded watchdog
Management	Telnet/Web/SSH/Console, Device Manager update, SNMP v1/v2c/v3 with TRAP
CPU / Memory / Flash	580MHz CPU, 128MB DDR2 RAM, 32MB SPI flash
Ethernet Interface	2×10/100Mbps RJ45 (WAN/LAN switchable)
Optional Interface	1×RS232 or 2×I/O (DI/DO configurable)
Power Input	DC 9~36V, over-current and reverse polarity protection
Operating Temperature	Normal: -35°C to +70°C Extended: -40°C to +75°C
Protection Level	IP30

2. Product Dimensions



The diagram shows the rear panel of the device with the following components from left to right: a BNC connector labeled 'BNC IN', a DC power input labeled 'DC IN' with a polarity symbol (minus on top, plus on bottom) and '5V 1A', a large central rectangular port labeled 'LAN IN' and 'LAN OUT', an audio output labeled 'AUDIO', and a BNC connector labeled 'BNC OUT'. A small label 'LAN IN' is also visible above the central port.

Note:

1. All dimensions are in millimeters (mm).
2. All dimensions are approximate and for reference only.
3. Dimensioned drawings are not intended for machining.
4. Dimensions are subject to part and manufacturing tolerances.
5. Specifications may change without prior notice.

3. Hardware Specifications

Category/Parameter	Specification
CPU and Storage	
CPU	580 MHz
RAM	128 MB DDR2
Flash	32 MB SPI
Connectivity and Interfaces	
Ethernet Ports	2 × 10/100 Mbps RJ45 (WAN/LAN switchable)
Power Interface	DC 9~36 V input, over-current and reverse-polarity protection, 2-pin industrial terminal
I/O Port	Optional 2 × I/O (DI/DO configurable)
Serial Port	Optional 1 × RS232
Reset Button	Pinhole reset button
SIM Slot	Dual SIM drawer slots
Antenna Connectors	4G: 1 × SMA (NA models: 2 × SMA); Wi-Fi: 1 × RP-SMA (optional)
Wi-Fi	
Radio Frequency	2.4 GHz (optional)



Category/Parameter	Specification
Max Transmission Bandwidth	Up to 150 Mbps
Transfer Protocol	IEEE 802.11 b/g/n
Transmit Power	802.11b:16dBm +/-2dBm(11Mbps) 802.11g:16dBm +/-2dBm(54Mbps) 802.11n @ 2.4GHz:16dBm +/-2dBm(HT20 MCS7) 802.11n @ 2.4GHz:16dBm +/-2dBm(HT40 MCS7)
Transmission Distance	50 meters by line of sight (Actual transmission distance depends on environment of the site.)
Power Rate	
Standby Power	80mA-90mA@12V
Working Power	100mA-120mA@12V
Peak Power	190mA@12V
Mechanical Specifications	
Product Dimensions	98.3 × 92 × 24 mm
Product Weight	259 g
Mounting Method	Panel or DIN-rail mounting
Protection Rating	IP30
Environment and Compliance	
Storage Temperature	-40°C to +85°C
Operating Temperature	Normal: -35°C to +70°C Extended:-40°C to +75°C
Ambient Humidity	5% to 95% RH (non-condensing)
Physical Characteristics	IEC60068-2-27, shockproof IEC60068-2-6, Vibration Resistance IEC60068-2-32, Free Fall
EMC Standard	EN61000-4-2, level 2, Static EN61000-4-3, level 2, Radiation Electric Field EN61000-4-4, level 2, Pulsed Electric Field EN61000-4-5, level 2, Surge EN61000-4-6, level 2, Conducted Disturbance Immunity Power Frequency EN61000-4-8, Magnetic Field Resistance, horizontal / vertical 400A/m (>level 2) Shock Wave Resistance, EN61000-4-12,level 2

Category/Parameter	Specification
Certifications	CE, CB, UKCA, E-MARK, FCC, IC, PTCRB, AT&T, Verizon, T-Mobile, RCM, CCC, EAC&FAC, UL, ANATEL, MIC&JATE, IEC62443, NOM, IFETEL

4. Software Specifications

Category/Parameter	Specification
Network Features	
Network Access	APN, VPDN
Access Authentication	CHAP, PAP
Network Type	GSM/GPRS/EDGE, UMTS/HSPA+/EVDO/TD-SCDMA, TDD LTE/FDD LTE (model dependent)
LAN protocol	ARP, Ethernet
WAN protocol	Static IP, DHCP, PPPoE
IP Applications	IPv4, Ping, Trace, DHCP Server/Relay/Client, DNS relay, DDNS, Telnet, IP passthrough
IP Routing	Static routing
NAT Functions	NAT
Security	
Network Security	SPI firewall, anti-DoS, ACL, URL filtering, port mapping, virtual IP mapping, IP-MAC binding
Data Security	PPTP, L2TP, GRE, IPSec, OpenVPN, WireGuard, ZeroTier
CA Certificates	Supported
Reliability	
Link Detection	Sends heartbeat packets to detect, auto redials when disconnected
Embedded Watchdog	Device runs self-detection, auto recovers from malfunctions
Hot Backup Mechanism	VRRP
Dual-SIM Switching	Supported
WLAN(Optional)	
Operating Mode	AP, Client (optional)

Category/Parameter	Specification
Security Features	WPA/WPA2, WEP/TKIP/AES
Intelligence	
DTU Function	TCP/UDP transparent transmission, TCP server mode, AT command support
Bridge	Modbus RTU to Modbus TCP
Network Management	
QoS Management	Traffic management
Configuration Methods	Telnet, Web, SSH, Console
Upgrade Methods	Web and Device Manager updates
Log	Local/remote logs with power-down retention
SMS Function	Status query, restart
Dial-on-demand	Dial-on-demand, data / SMS activation
Network Management Function	Device Manager, batch management
SNMP Function	SNMP v1/v2c/v3 with TRAP
Traffic Management	Supports data traffic threshold, traffic statistics and traffic alarm
Alarm	System restart alarm, LAN port online/offline alarm, data traffic alarm, SIM card failure alarm ,etc.
Maintenance Tools	Ping, route tracking, network speed test
Status Query	System status, modem status, network connection status, and routing status

5. Ordering Information

Model Rule

Model code: IR302-<WMNN>-<WLAN/NA>-<IO/S/NA>

<WMNN>: Cellular Type & Module

<WLAN/NA>: Wi-Fi option

<IO/S/NA>: Interface option (IO =2×IO, S =1×RS232, NA =no optional interface)

Model List

Model Pattern	Region	Network Type	<WMNN>: Cellular Type & Module	<WLAN/NA>	<IO/S/NA>
IR302-FQ58- <WLAN/NA>- <IO/S/NA>	Europe & APAC	LTE CAT4	FDD B1/B3/B7/B8/B20/B28A; WCDMA B1/B8; GSM B3/B8	WLAN or NA	IO / S / NA
IR302-FQ58- <WLAN/NA>- <IO/S/NA>-L	Europe, APAC, Australia	LTE CAT4	FDD B1/B3/B5/B7/B8/B20/B28; TDD B38/B40/B41; WCDMA B1/B5/B8; GSM/EDGE B3/B8	WLAN or NA	IO / S / NA
IR302-FQ53- <WLAN/NA>- <IO/S/NA>	EMEA	LTE CAT1	FDD B1/B3/B7/B8/B20/B28; WCDMA B1/B8; GSM/EDGE B3/B8	WLAN or NA	IO / S / NA
IR302-FQ38- <WLAN/NA>- <IO/S>	North America (AT&T/ Verizon)	LTE CAT4	FDD B2/B4/B5/B12/B13/B14/B66/B71; WCDMA B2/B4/B5	WLAN or NA	IO / S
IR302-FQ33- <WLAN/NA>- <IO/S>	North America (AT&T/ Verizon)	LTE CAT1	FDD B2/B4/B5/B12/B13/B25/B26; WCDMA B2/B4/B5	WLAN or NA	IO / S
IR302-FQ68- <WLAN/NA>- <IO/S>	Latin America	LTE CAT4	FDD B1/B2/B3/B4/B5/B7/B8/B28/B66; TDD B40; WCDMA B1/B2/B4/B5/B8; GSM B2/B3/B5/B8	WLAN or NA	IO / S
IR302-FQ88- <WLAN/NA>-S	Japan	LTE CAT4	FDD B1/B3/B8/B18/B19/B26; TDD B41; WCDMA B1/B6/B8/B19	WLAN or NA	S
IR302-EN00- WLAN	Global	NA	No cellular module	WLAN	NA
IR302-LQ28- <WLAN/NA>- <S/NA>	China	LTE CAT4	FDD B1/B3/B5/B8; TDD B34/B38/B39/B40/B41; WCDMA B1/B8; TD-SCDMA B34/B39; EVDO/CDMA BC0; GSM B3/B8	WLAN or NA	S / NA
IR302-LQ28- <WLAN/NA>- <S/NA>-L	China	LTE CAT4	FDD B1/B3/B5/B8; TDD B34/B38/B39/B40/B41; WCDMA B1/B5/B8; GSM/EDGE B3/B8	WLAN or NA	S / NA



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